Track 5

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Problem Statement:

A Platform to Track Illicit Crypto Flow, Flag Fraudulent Accounts, and Analyze Suspicious Financial Transactions

Explanation:

Cryptocurrency adoption has grown rapidly, and with it, the scale of illicit activity moving through digital assets. According to Chainalysis' 2026 Crypto Crime Report, illicit cryptocurrency addresses received at least $154 billion in 2025, a 162% year-over-year increase, driven by industrialized scam operations, nation-state actors, and increasingly sophisticated laundering networks. Fraud and impersonation scams alone accounted for an estimated $17 billion in losses, with AI-enabled social-engineering tactics now overtaking traditional cyberattacks as the leading method criminals use to steal funds. The pseudonymous, borderless, and fast-moving nature of blockchain transactions makes it difficult for exchanges, financial institutions, and regulators to trace fund origins, identify fraudulent accounts, and flag suspicious activity before losses occur. Existing compliance workflows often depend on manual review and fragmented data across chains and platforms, resulting in delayed detection and limited visibility into complex laundering structures. There is a growing need for an intelligent platform capable of continuously monitoring on-chain and off-chain activity, correlating identifiers across sources, and generating timely, actionable alerts to support investigators and compliance teams while operating within applicable legal and regulatory frameworks.

Objectives:

- To assist financial institutions, exchanges, and regulatory agencies in the early detection of illicit cryptocurrency flows.
- To enhance financial intelligence by integrating and correlating information from multiple lawful on-chain and off-chain data sources.
- To support investigators and compliance analysts in identifying fraudulent accounts, suspicious wallets, and potential laundering networks.
- To improve the speed, accuracy, and efficiency of financial crime investigations through automated intelligence generation.
- To support evidence-based investigation by providing explainable risk indicators and transaction insights for suspicious cryptocurrency activity.

Expected Features:

1. **Multi-Source Data Collection**

   Collect and integrate information from blockchain explorers, cryptocurrency exchanges (where legally accessible), banking transaction records, suspicious transaction reports (STRs), sanctions databases, public blockchain datasets, and open-source intelligence (OSINT).

2. **Intelligent Entity Correlation**

   Identify relationships between wallet addresses, exchange accounts, and other digital identifiers to help investigators uncover fraud rings and laundering networks, drawing on red-flag indicator patterns published by FATF.

3. **Suspicious Transaction Detection**

   Automatically detect **financial anomaly**, unusual transaction patterns, rapid fund layering, mixer/tumbler usage, structuring, and other behaviors that may indicate financial crime.

4. **Interactive Intelligence Dashboard**

   Give investigators and compliance teams a centralized dashboard to monitor alerts, visualize transaction trends, search records, and manage ongoing cases.

5. **Automated Alert Generation**

   Notify investigators whenever predefined risk indicators, such as high-risk wallet interactions or abnormal transaction velocity, are detected.

6. **Search & Investigation Support**

   Allow advanced searches using wallet addresses, transaction hashes, or account identifiers, while maintaining historical records for future reference.

7. **Reporting & Evidence Management**

   Generate structured intelligence reports, maintain investigation logs, and support secure documentation of financial evidence.

8. **Security & Access Control**

   Ensure secure authentication, role-based access, audit logging, and protection of sensitive financial and investigative information.

9. **Scalability**

   Support continuous monitoring of large volumes of heterogeneous blockchain and financial data, with the ability to integrate additional intelligence sources over time.

Tech Stack:

Participants are encouraged to justify their technology choices based on scalability, security, maintainability, performance, and suitability for the proposed solution rather than adopting any predefined stack.

Resources:

[1] U.S. Department of Justice, "Two Charged in Connection With Cryptocurrency Money Laundering Service That Allegedly Laundered Over $389 Million in Unlawful Transactions," *U.S. Attorney's Office, Eastern District of Pennsylvania*, Jun. 11, 2026. [Online]. Available: https://www.justice.gov/usao-edpa/pr/two-charged-connection-cryptocurrency-money-laundering-service-allegedly-laundered. [Accessed: Jul. 17, 2026].

[2] Financial Action Task Force (FATF), *Virtual Assets: Red Flag Indicators of Money Laundering and Terrorist Financing*. Paris, France: FATF, Sep. 2020. [Online]. Available: https://www.fatf-gafi.org/en/publications/Methodsandtrends/Virtual-assets-red-flag-indicators.html. [Accessed: Jul. 17, 2026].

[3] Chainalysis, *The 2026 Crypto Crime Report*. New York, NY, USA: Chainalysis, 2026. [Online]. Available: https://www.chainalysis.com/reports/crypto-crime-2026/. [Accessed: Jul. 17, 2026].

[4] S. Meiklejohn, M. Pomarole, G. Jordan, K. Levchenko, D. McCoy, G. M. Voelker, and S. Savage, "A Fistful of Bitcoins: Characterizing Payments Among Men with No Names," in *Proc. ACM Internet Measurement Conf. (IMC)*, Barcelona, Spain, Oct. 2013, pp. 127–140, doi: 10.1145/2504730.2504747.

[5] F. Reid and M. Harrigan, "An Analysis of Anonymity in the Bitcoin System," in *Security and Privacy in Social Networks*, Y. Altshuler, Y. Elovici, A. B. Cremers, N. Aharony, and A. Pentland, Eds. New York, NY, USA: Springer, 2013, pp. 197–223, doi: 10.1007/978-1-4614-4139-7_10.

[6] S. Nakamoto, *Bitcoin: A Peer-to-Peer Electronic Cash System*, 2008. [Online]. Available: https://bitcoin.org/bitcoin.pdf. [Accessed: Jul. 17, 2026].